

2019



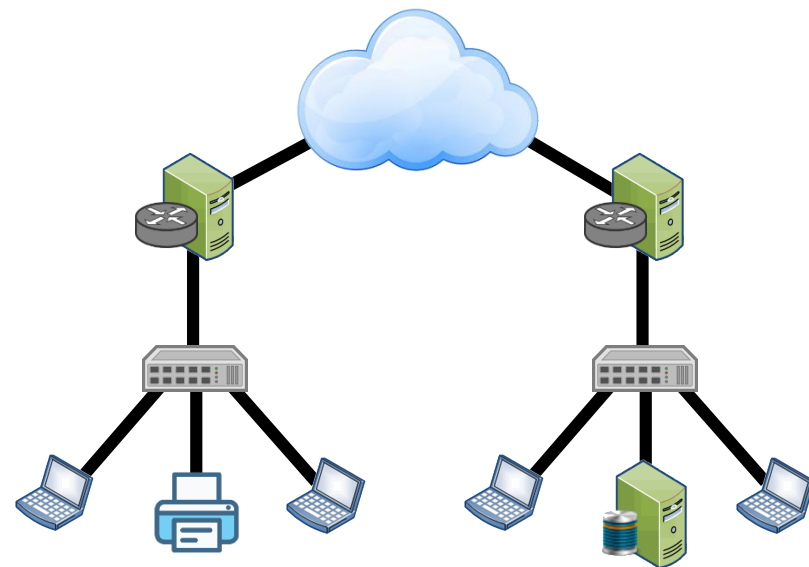
IT Basics

Bus Basics B1

General

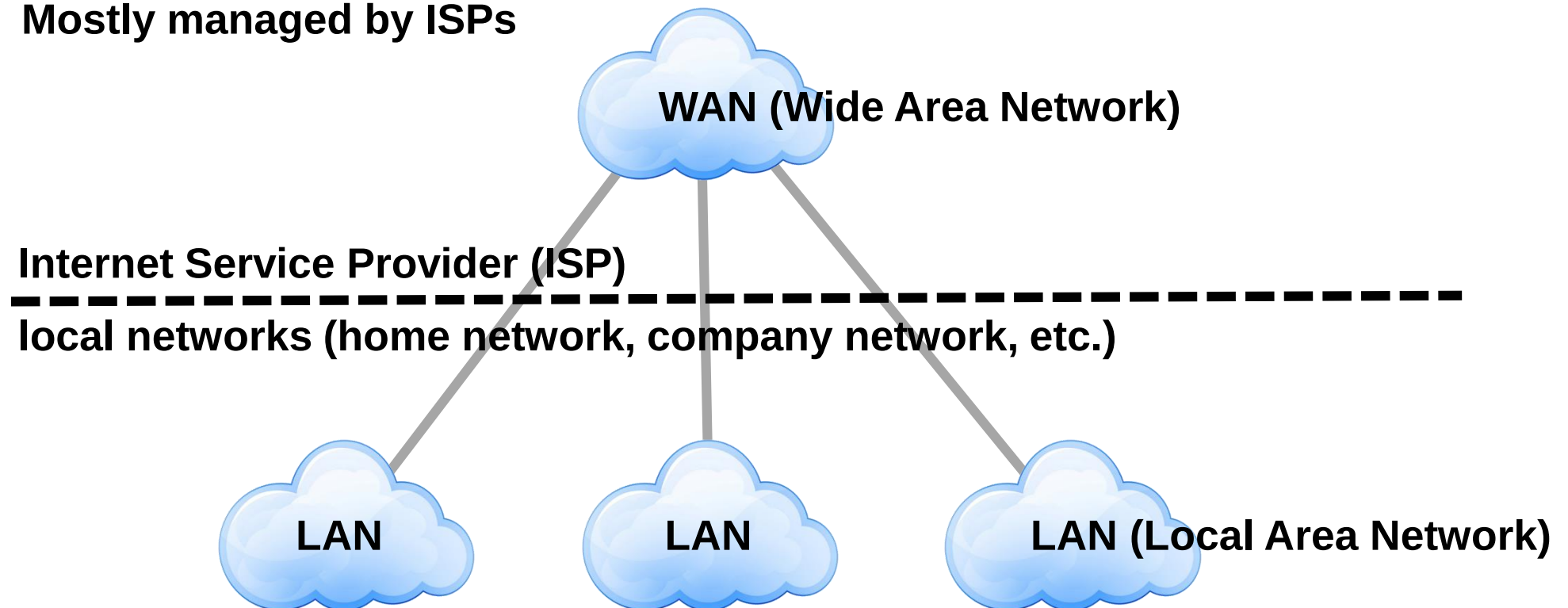
What is a network

- A network consists of two to several million devices
- Sharing resources (printers, memory, etc.)
- Services (communication, IoT, etc.)
- Internet is the largest existing network and is a network of private and public networks



Network infrastructures

- Long distances
- LAN spanning
- Mostly managed by ISPs



- short distances
- e.g. home network, company network; often managed locally

- **Network addresses serve as unique identifiers in the network**
- **Each device in the network has two different addresses**
 - Physical address -> MAC address
 - Logical address -> IP address
- **Link between IP and MAC address (ARP)**

- **Unique identification**
- **Is allocated by Manufacture of Internet-enabled devices**
- **The MAC address can be used to identify some of the manufacturers.**
- **Address is 48 bit long**
 - Example: 00-16-21-25-AF-D7

IP-Address (IPv4)

- **Internet Protocol**
- **Temporary identification in networks**
- **Public (global) & private IP addresses**
- **Public IP addresses**
 - are administered/allocated by IANA (Internet Assigned Numbers Authority).
 - are unique worldwide
 - are directly accessible on the Internet
- **Private IP addresses**
 - are mostly managed/assigned locally
- **Adresse ist 32 Bit lang**
 - Example: 192.168.0.10

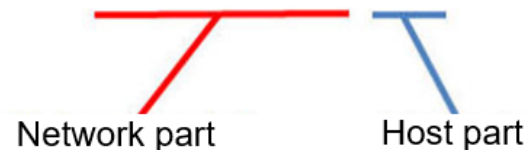


- Loopback
 - 127.0.0.0/8
- RFC1918 - Private Address Space
 - 10.0.0.0/8, 172.16.0.0/12 und 192.168.0.0/16
- RFC3927 - APIPA / Link-Local
 - 169.254.0.0/16
- RFC6598 - Shared Address Space (CGN)
 - 100.64.0.0/12
- RFC5737 - Documentation prefixes
 - 192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24
- RFC8190 - Special-Purpose IP Address Registries
 - Complete list of special prefixes

IP address structure (IPv4)

- **Dotted Decimal Notation x.x.x.x**
 - Each octet (x) has 8 bits -> range 0-255
- **IP address is divided into a net ID and a host ID using the subnet mask**
- **Example:** Subnet mask 255.255.255.0

IP-Address 192.168.178.125



Subnet mask: The first 3 number blocks

Network ID: 192.168.178.

Host ID: 125

The network includes: 192.168.178.1 – 192.168.178.254

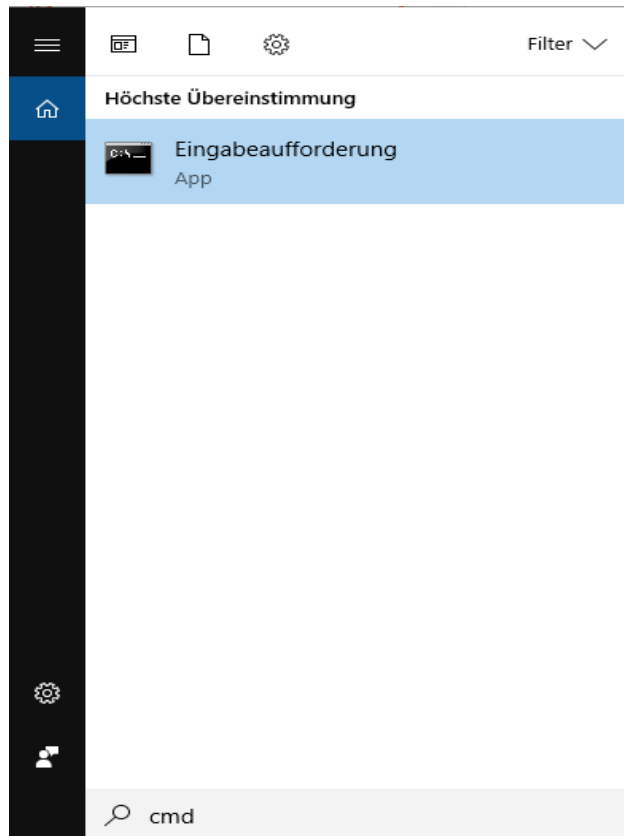
Binary system:

<https://www.youtube.com/watch?v=LpuPe81bc2w&t=252s> (1'50'')

MAC-, IP-Adresse & Subnet mask

Check your IP Settings

-> *ipconfig /all*



```
Eingabeaufforderung
C:\Users\ottbe>ipconfig/all

Windows-IP-Konfiguration

Hostname . . . . . : L-OTTBE
Primäres DNS-Suffix . . . . . : eu.belimonet.com
Knotentyp . . . . . : Hybrid
IP-Routing aktiviert . . . . . : Nein
WINS-Proxy aktiviert . . . . . : Nein
DNS-Suffixsuchliste . . . . . : eu.belimonet.com
                                sap.belimonet.com
                                na.belimonet.com
                                asia.belimonet.com
                                belimonet.com

Ethernet-Adapter Ethernet 2:

Medienstatus. . . . . : Medium getrennt
Verbindungsspezifisches DNS-Suffix:
Beschreibung. . . . . : Fortinet SSL VPN Virtual Ethernet Adapter
Physische Adresse . . . . . : 00-09-0F-AA-00-01
DHCP aktiviert. . . . . : Ja
Autokonfiguration aktiviert . . . : Ja

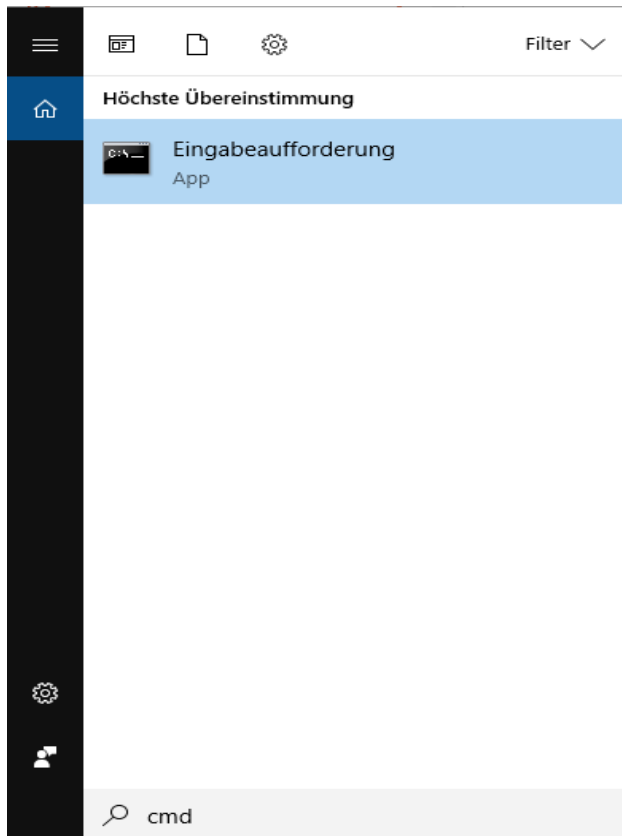
Ethernet-Adapter Ethernet:

Verbindungsspezifisches DNS-Suffix: eu.belimonet.com
Beschreibung. . . . . : Intel(R) Ethernet Connection I218-LM
Physische Adresse . . . . . : D0-BF-9C-E2-75-F2
DHCP aktiviert. . . . . : Ja
Autokonfiguration aktiviert . . . : Ja
Verbindungslokale IPv6-Adresse . . : fe80::20d0:6fcb:3912:8859%22(Bevorzugt)
IPv4-Adresse . . . . . : 10.20.14.69(Bevorzugt)
Subnetzmaske . . . . . : 255.255.0.0
Lease erhalten. . . . . : Dienstag, 25. Juni 2019 06:59:40
Lease läuft ab. . . . . : Samstag, 29. Juni 2019 06:59:40
Standardgateway . . . . . : 10.20.0.1
DHCP-Server . . . . . : 10.10.100.92
DHCPv6-IAID . . . . . : 97566620
DHCPv6-Client-DUID. . . . . : 00-01-00-01-20-F7-CC-4A-D0-BF-9C-E2-75-F2
DNS-Server . . . . . : 10.10.100.90
                        10.10.100.92
NetBIOS über TCP/IP . . . . . : Aktiviert
```

MAC-, IP-Adresse & Subnet mask

Check you ARP list

-> arp -a



```
C:\Users\ottbe>arp -a

Schnittstelle: 192.168.0.115 --- 0xd
  Internetadresse    Physische Adresse    Typ
  192.168.0.1        34-2c-c4-68-7e-ec    dynamisch
  192.168.0.255      ff-ff-ff-ff-ff-ff    statisch
  224.0.0.2          01-00-5e-00-00-02    statisch
  224.0.0.22         01-00-5e-00-00-16    statisch
  224.0.0.251        01-00-5e-00-00-fb    statisch
  224.0.0.252        01-00-5e-00-00-fc    statisch
  239.255.255.250    01-00-5e-7f-ff-fa    statisch
  255.255.255.255    ff-ff-ff-ff-ff-ff    statisch

Schnittstelle: 10.9.1.11 --- 0x12
  Internetadresse    Physische Adresse    Typ
  10.9.1.12          00-09-0f-aa-00-02    dynamisch
  224.0.0.2          01-00-5e-00-00-02    statisch
  224.0.0.22         01-00-5e-00-00-16    statisch
  224.0.0.251        01-00-5e-00-00-fb    statisch
  224.0.0.252        01-00-5e-00-00-fc    statisch
  239.255.255.250    01-00-5e-7f-ff-fa    statisch
  255.255.255.255    ff-ff-ff-ff-ff-ff    statisch
```

- Ports are part of every IP communication
- Valid port numbers are 0...65'535
- System Ports
- User Ports
- Dynamic Ports

Port	TCP	UDP	Beschreibung	Status
0		UDP	reserviert	offiziell
1	TCP	UDP	TCP Port Service Multiplexer (TCPMUX)	offiziell
2	TCP	UDP	CompressNET ^[1] Management Utility ^[2]	offiziell
3	TCP	UDP	CompressNET ^[1] Compression Process ^[3]	offiziell
4	TCP	UDP	Mesh Connection Protocol	inoffiziell
5	TCP	UDP	Remote Job Entry	offiziell
6	UDP	UDP	nicht zugewiesen	offiziell
7	TCP	UDP	Echo	offiziell
8	TCP	UDP	nicht zugewiesen	offiziell
9	TCP	UDP	Discard	offiziell
10	TCP	UDP	nicht zugewiesen	offiziell
11	TCP	UDP	Systeminformationen über den (systat-Service ^[4])	offiziell
12	TCP	UDP	LBBP	inoffiziell
13	TCP	UDP	Daytime (RFC 867 ^[5])	offiziell
14	TCP	UDP	nicht zugewiesen	offiziell
15	TCP	UDP	netstat Service ^[4]	inoffiziell
16	TCP	UDP	nicht zugewiesen	offiziell
17	TCP	UDP	Quote of the Day (QOTD) – Protokoll, um eine kurze „Nachricht des Tages“ zu übermitteln	offiziell
18	TCP	UDP	Message Send Protocol – Protokoll, um Textnachrichten zwischen Computern zu übermitteln	offiziell
19	TCP	UDP	Character Generator Protocol (CHARGEN)	offiziell

44818	TCP	UDP	EtherNet/IP (Industrial Ethernet)	offiziell
47808	TCP	UDP	BACnet	offiziell
49000	TCP	UDP	Server für Statusinformationen, SOAP (z. B. bei Fritz!Box)	inoffiziell
49151	TCP	UDP	reserviert	offiziell

Recap

- MAC Adresse
- IP Adresse
- Netz-ID / Host-ID
- Port

Useful links to recap at home:

<https://www.youtube.com/watch?v=ddM9AcreVqY> (14'48'')

<https://www.youtube.com/watch?v=XQ3T14SIIV4> (20'17'')

https://www.youtube.com/watch?v=4xlzlgYGqW8&list=PLI9NdZbdtA0wK8OIgttkScKRxcMkvoev_&index=3 (10'58'')

Network Components

Client and Server

- All end devices in the network are classified as hosts
- Servers provide services (e.g. web server)
- Client use services (e.g. web browser)



Webserver



Mailserver



Fileserver



Tablet



Computer



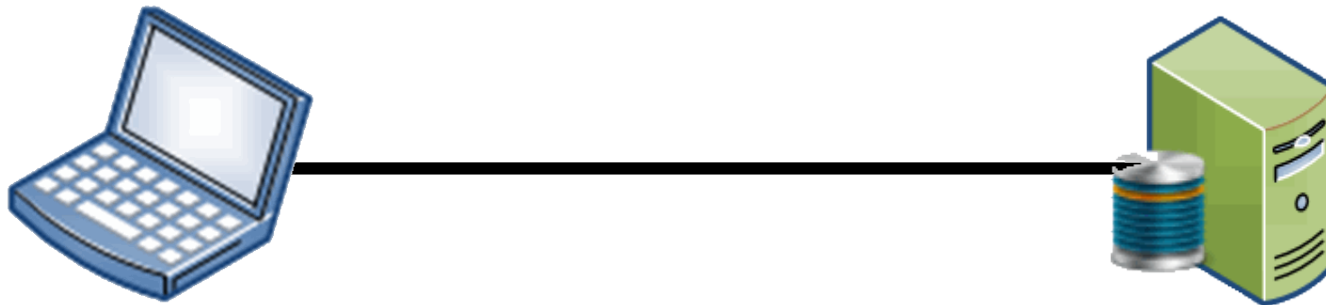
Printer



Laptop

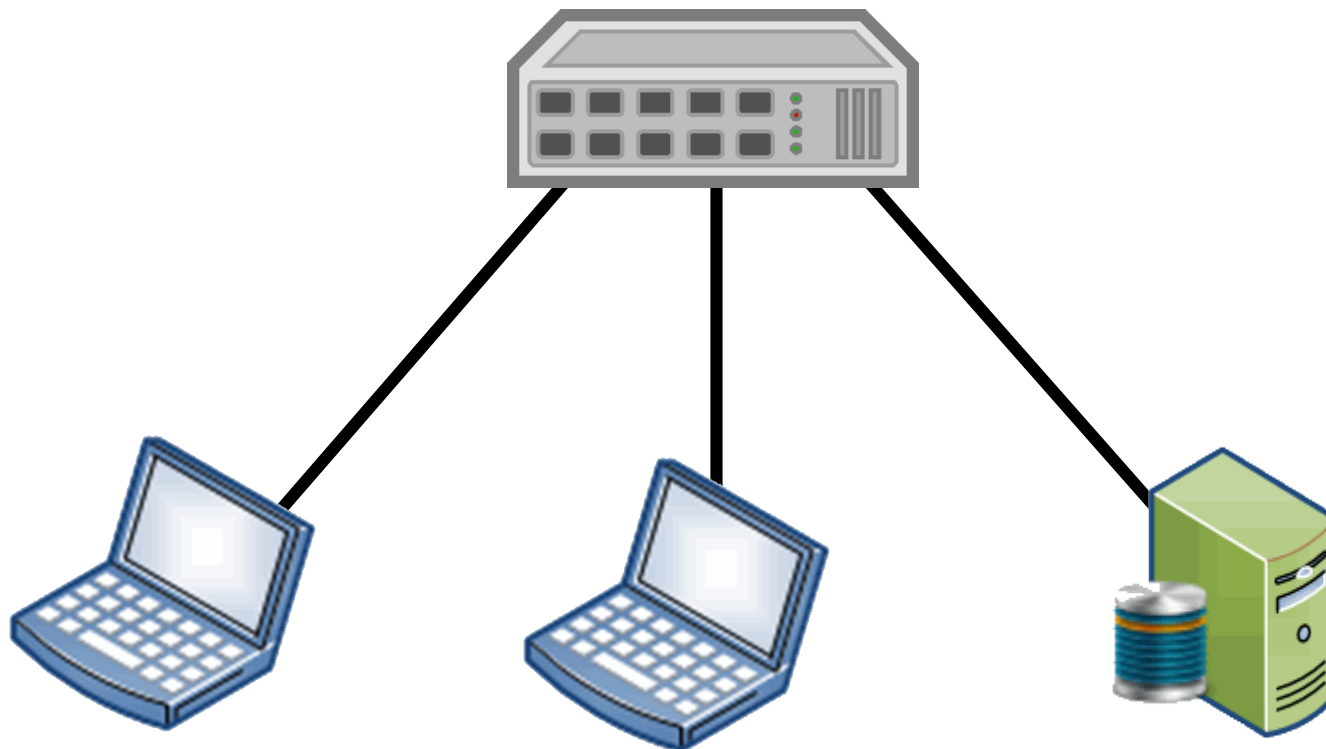
Peer to Peer

- e.g. Access local Energy Valve webserver with a laptop



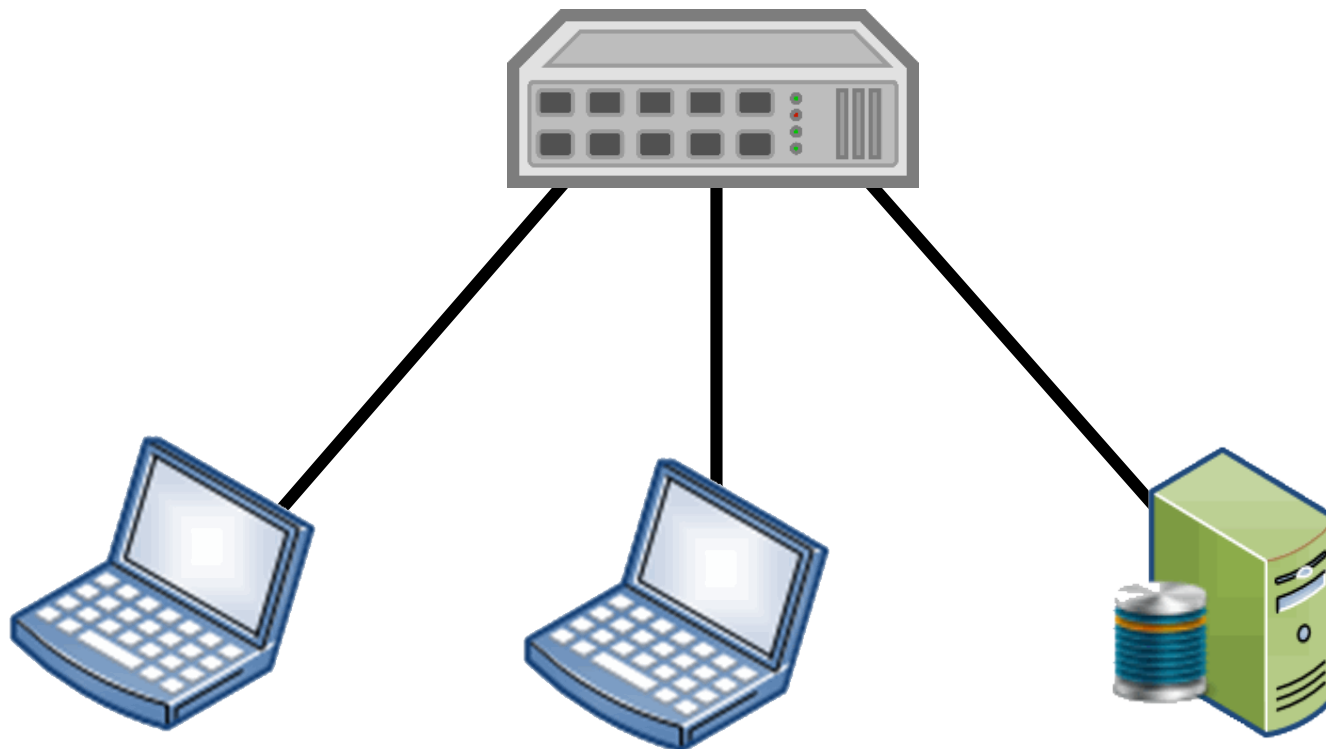
Hub (out of date)

- **Connects end devices to each other**
- **Sends all incoming frames to all other devices**



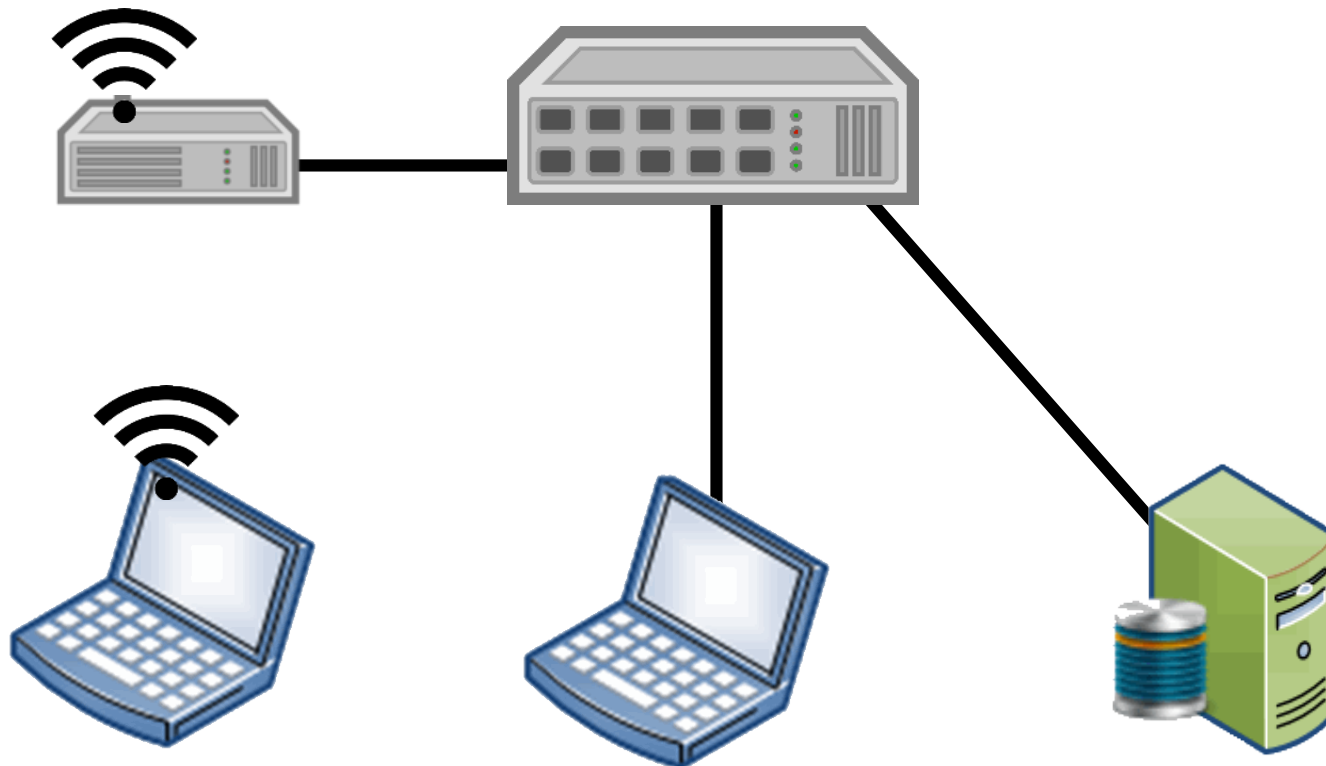
Switch

- **Connects end devices to each other**
- **Forwards incoming frames to the device addressed**
- **Manages a list with internal identifier**
- **Data Link Layer**



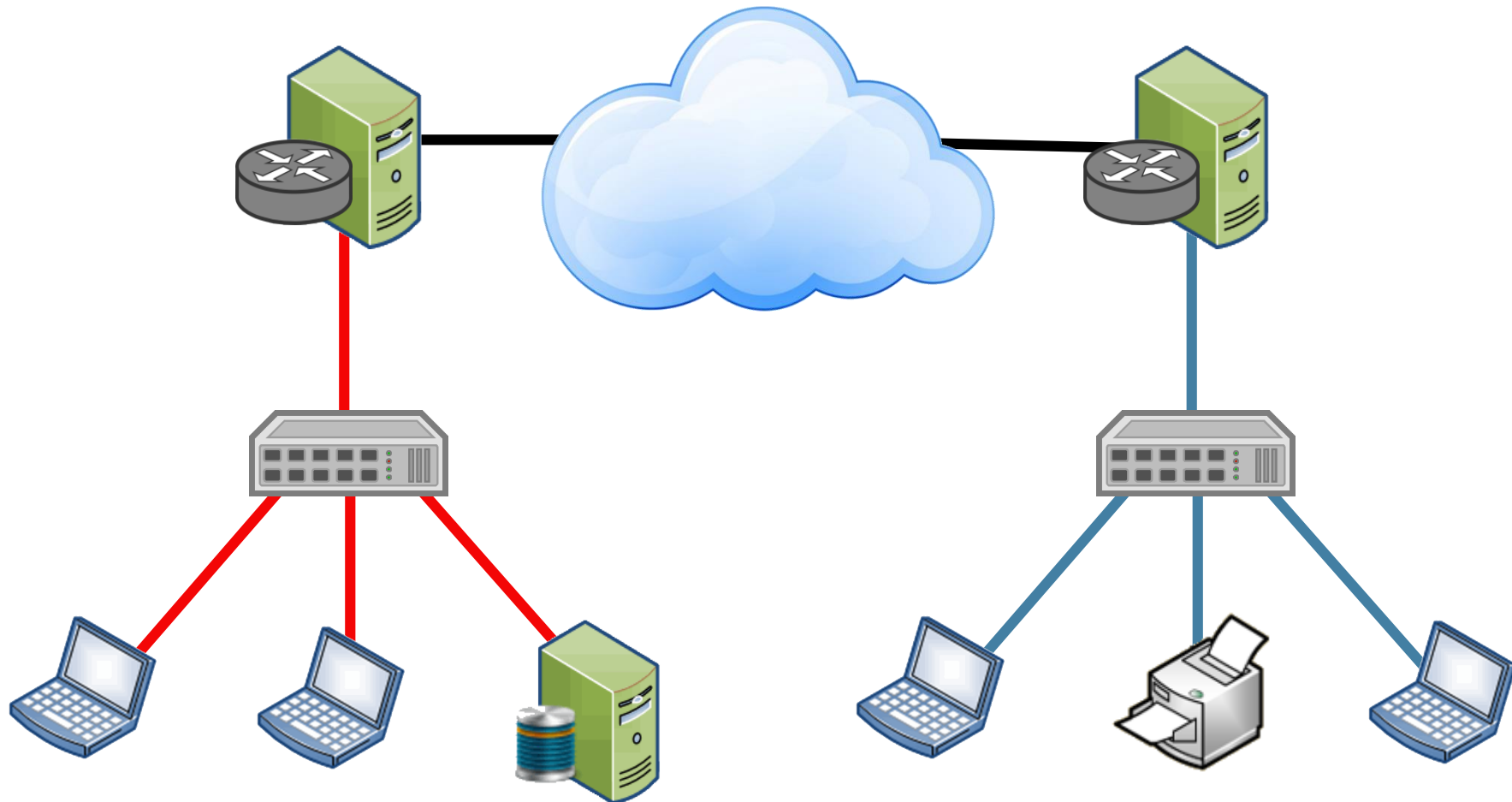
Wi-Fi Access Point

- **WLAN**

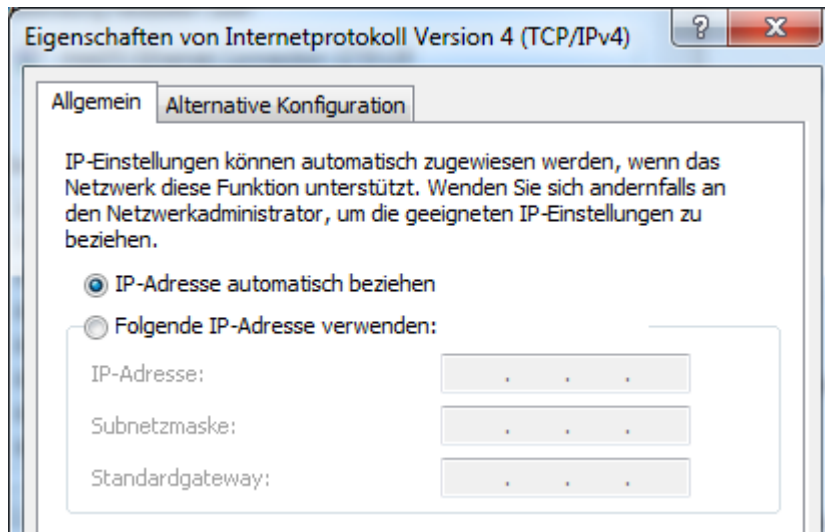


Router

- **Connects networks (logical network, LAN and WAN) by routing data packets to other networks**
- **Allocates IP addresses (DHCP server)**



Network settings

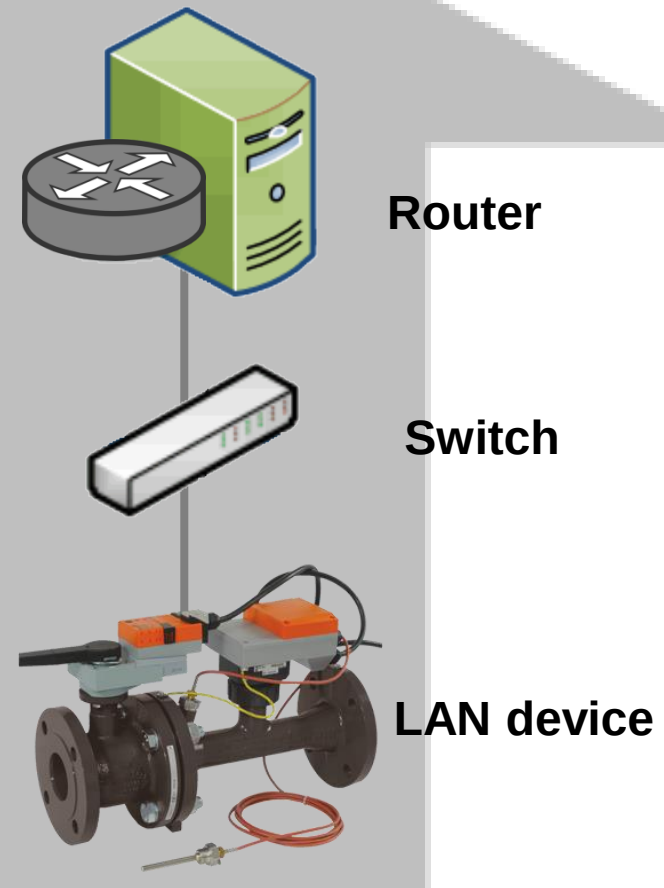


Static:

IP address and network parameters must be set manually. Must match the router settings (IP range, subnet mask, etc.).

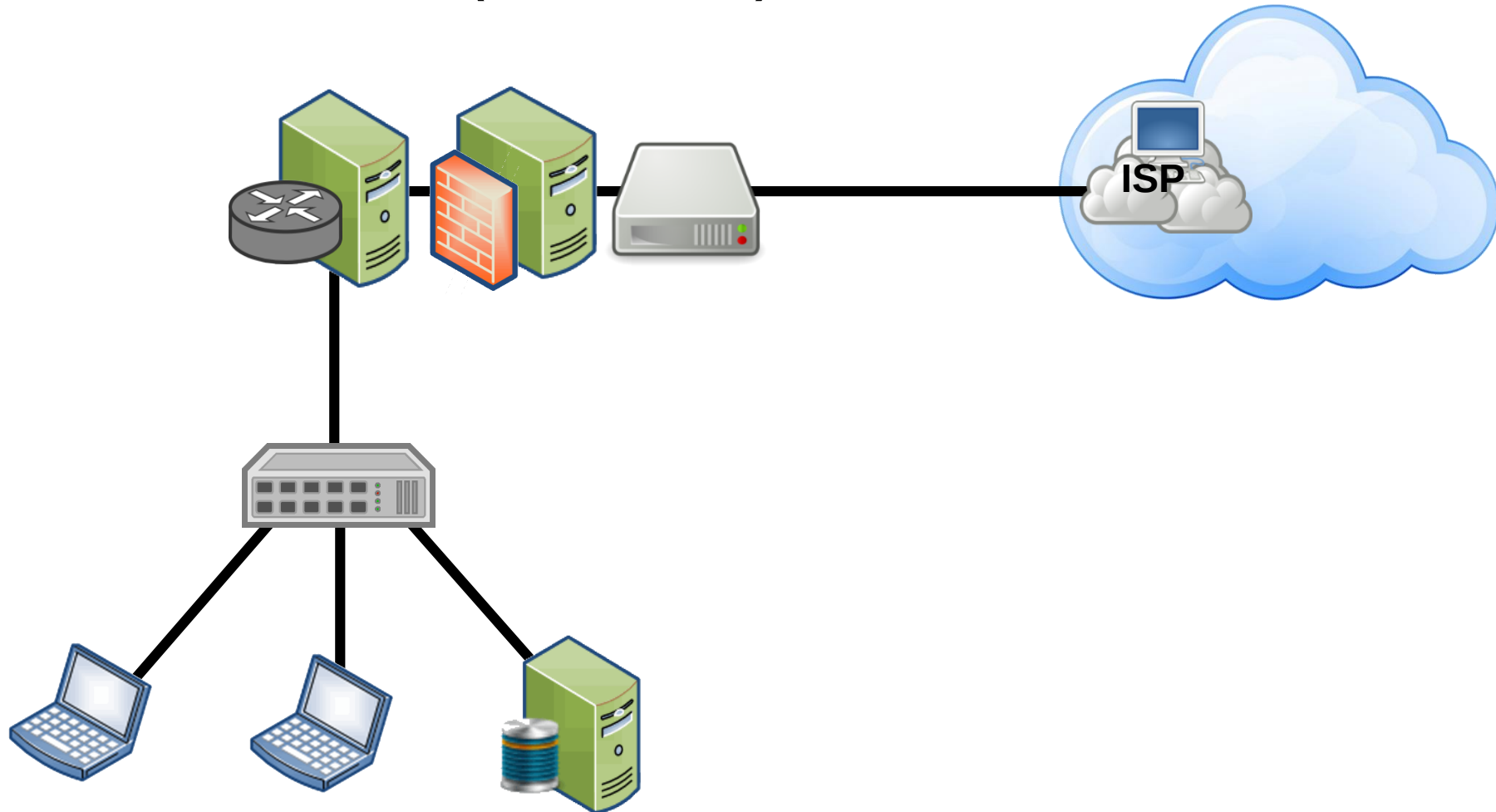
Dynamic (DHCP):

The device acts as a DHCP client and requests an IP address and all network parameters from the DHCP server (e.g. router).

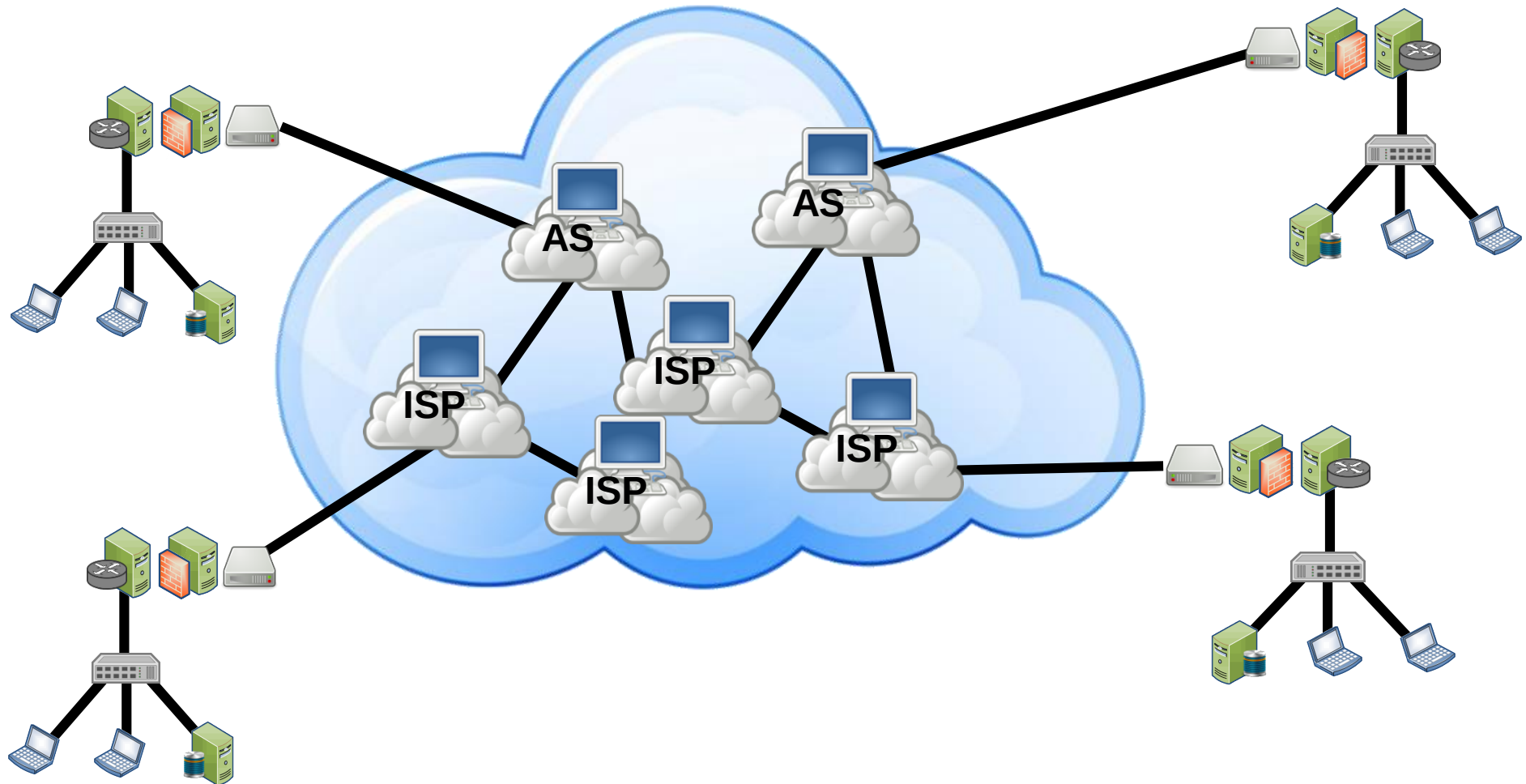


Internet Service Provider (ISP)

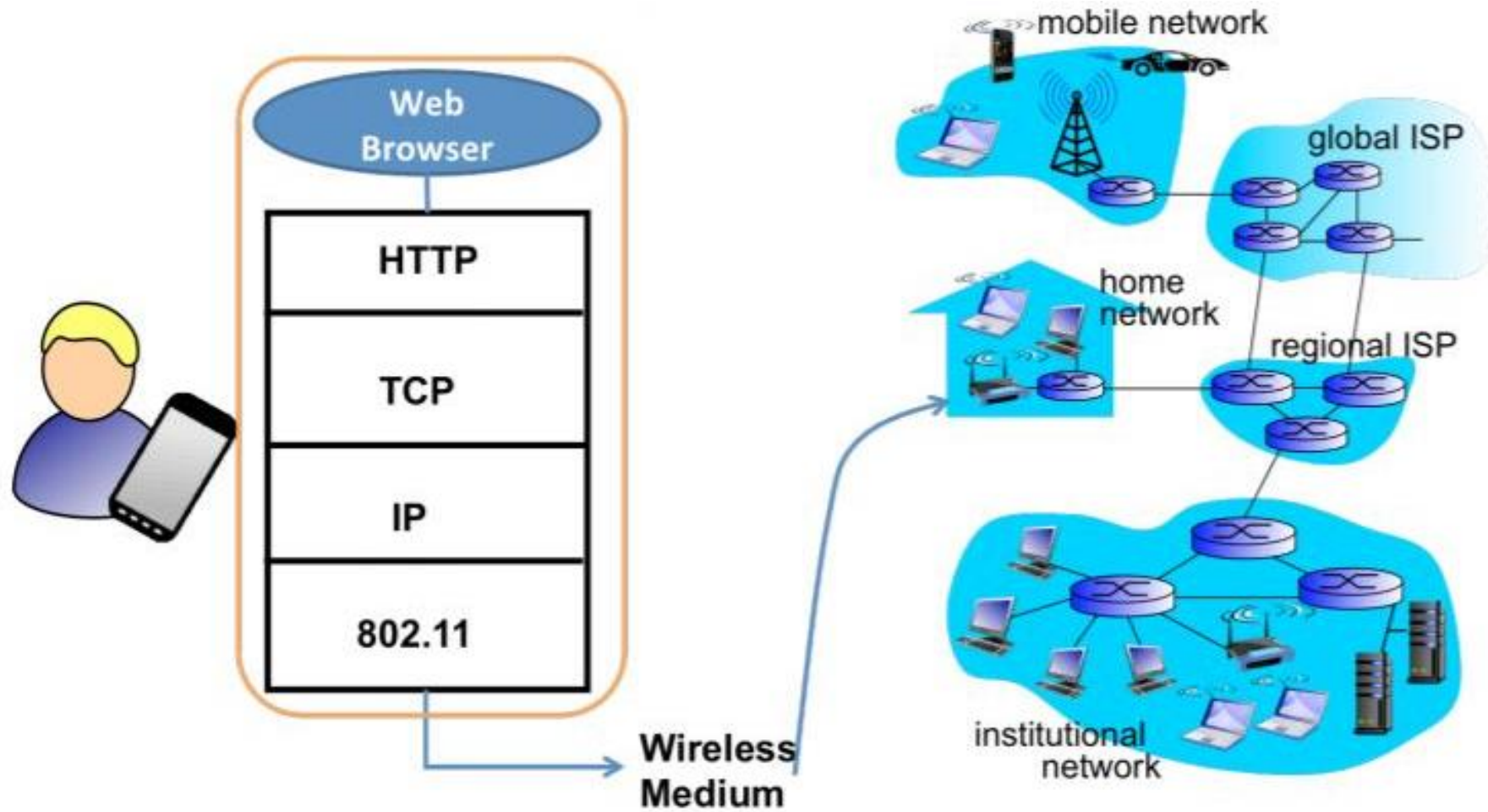
- **Service provider for connection to a WAN**
- **Allocates IP addresses (DHCP server)**



- Networking of individual entities ISP and AS (autonomous system)



Internet

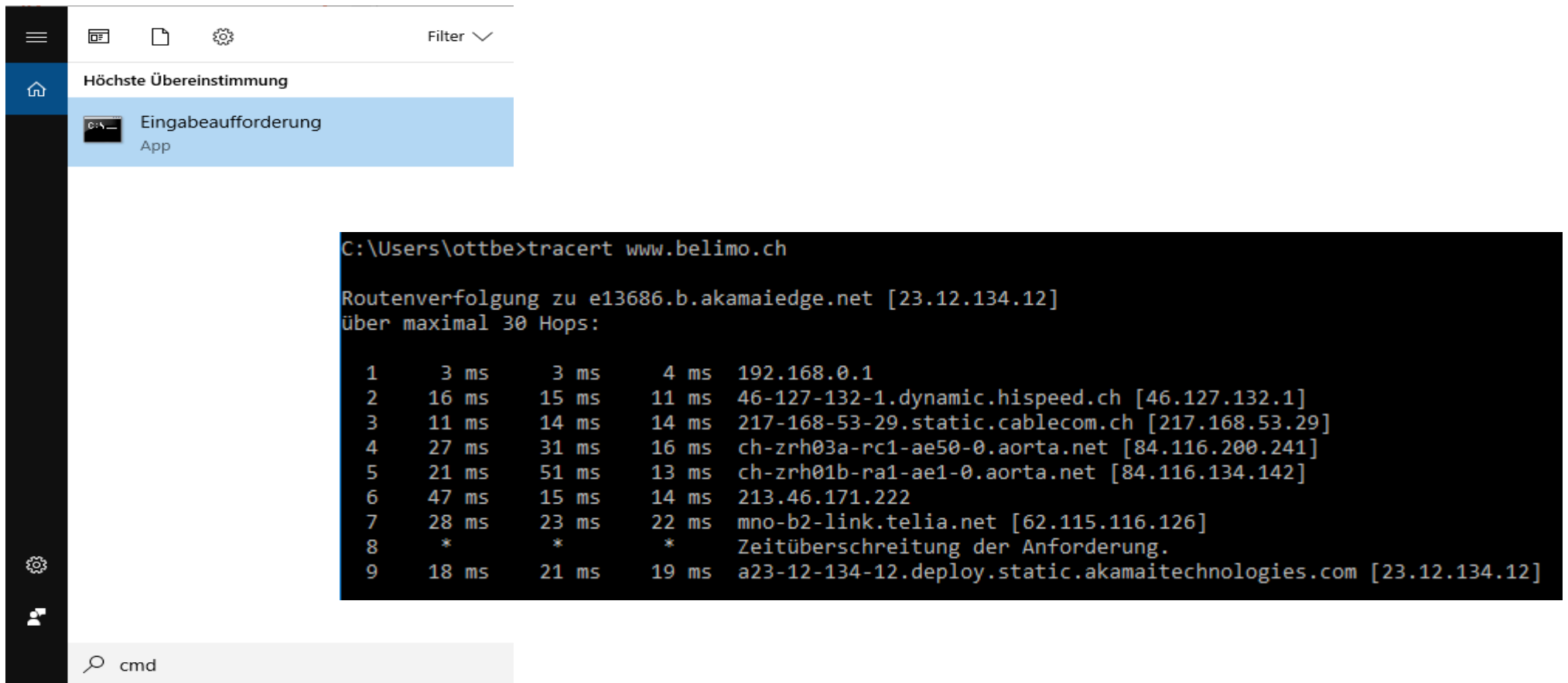


Öffentliche IP-Adresse (IPv4)

Check:

<https://whatismyipaddress.com>

***Check route to www.belimo.ch
-> tracert www.belimo.ch***



The screenshot shows a Windows Start menu on the left with the search bar containing 'cmd'. The 'Eingabeaufforderung' (Command Prompt) app is highlighted. To the right, a black Command Prompt window displays the output of the command 'tracert www.belimo.ch'. The output shows the route from the local host to the destination IP 23.12.134.12, passing through several hops with associated RTT times.

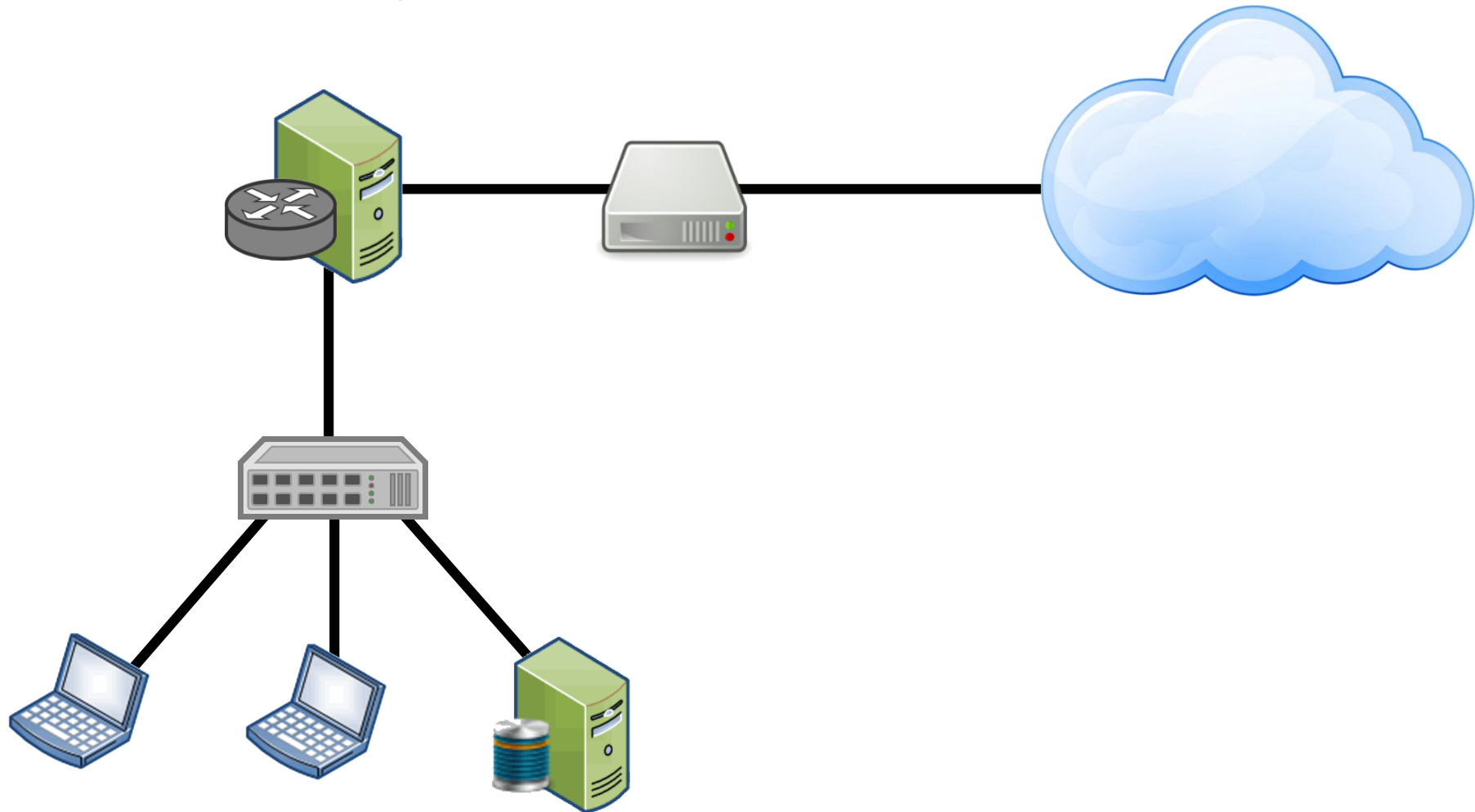
```
C:\Users\ottbe>tracert www.belimo.ch

Routenverfolgung zu e13686.b.akamaiedge.net [23.12.134.12]
über maximal 30 Hops:

 1    3 ms    3 ms    4 ms  192.168.0.1
 2   16 ms   15 ms   11 ms  46-127-132-1.dynamic.hispeed.ch [46.127.132.1]
 3   11 ms   14 ms   14 ms  217-168-53-29.static.cablecom.ch [217.168.53.29]
 4   27 ms   31 ms   16 ms  ch-zrh03a-rc1-ae50-0.aorta.net [84.116.200.241]
 5   21 ms   51 ms   13 ms  ch-zrh01b-ra1-ae1-0.aorta.net [84.116.134.142]
 6   47 ms   15 ms   14 ms  213.46.171.222
 7   28 ms   23 ms   22 ms  mno-b2-link.telia.net [62.115.116.126]
 8    *      *      *      Zeitüberschreitung der Anforderung.
 9   18 ms   21 ms   19 ms  a23-12-134-12.deploy.static.akamaitechnologies.com [23.12.134.12]
```

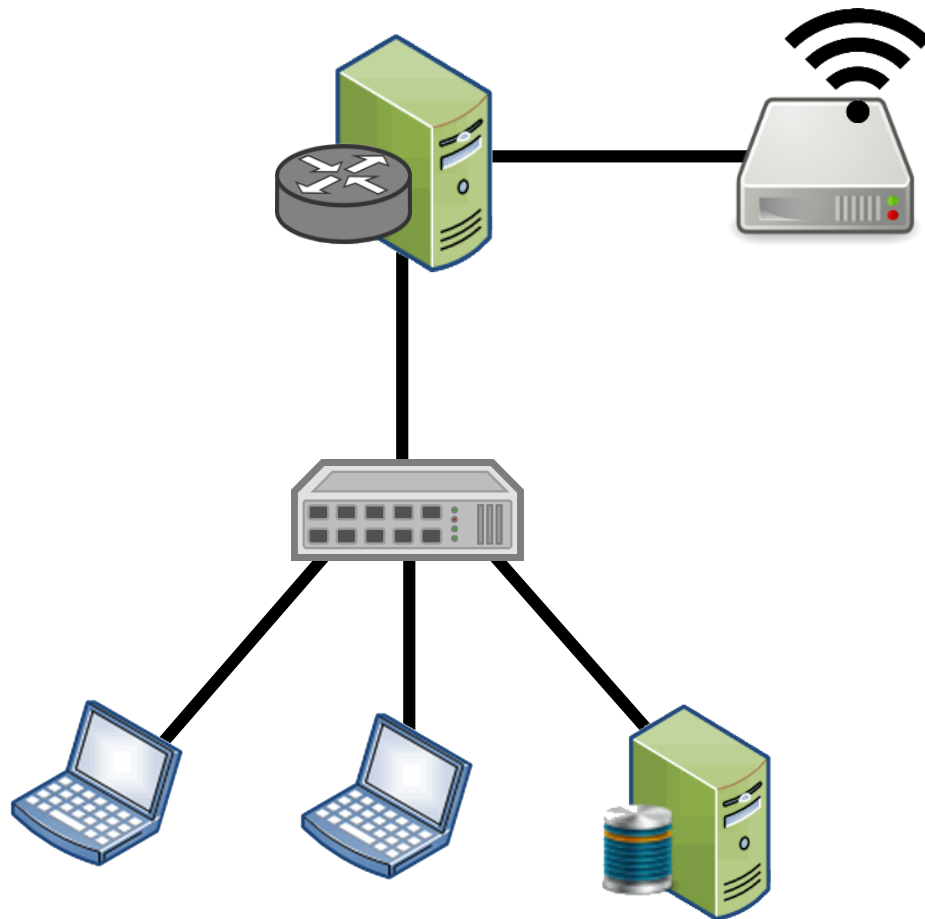
Modem

- **Signal conversion**
 - Ethernet <-> fibreglass



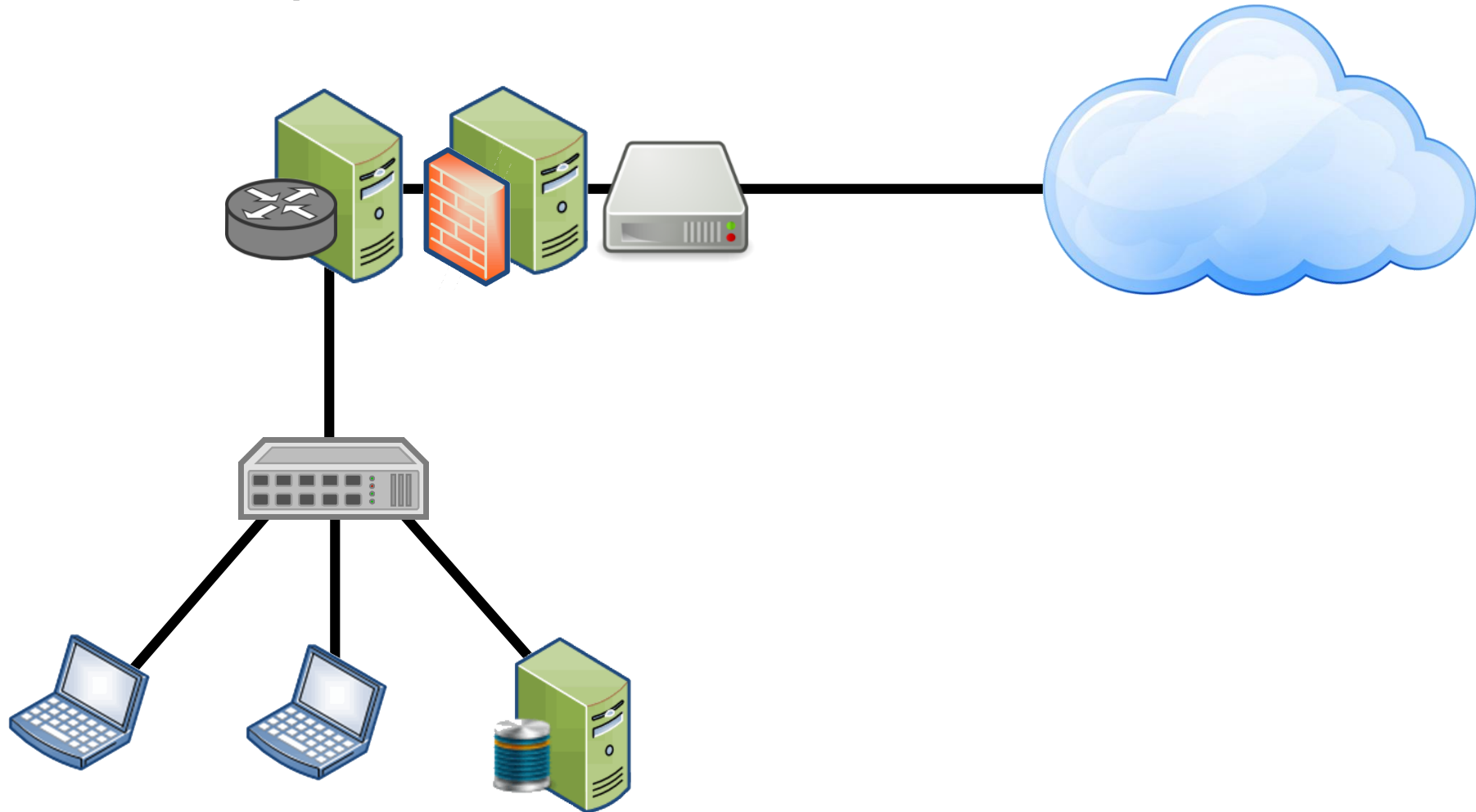
Modem

- **Signal conversion**
 - Ethernet <-> Mobile network



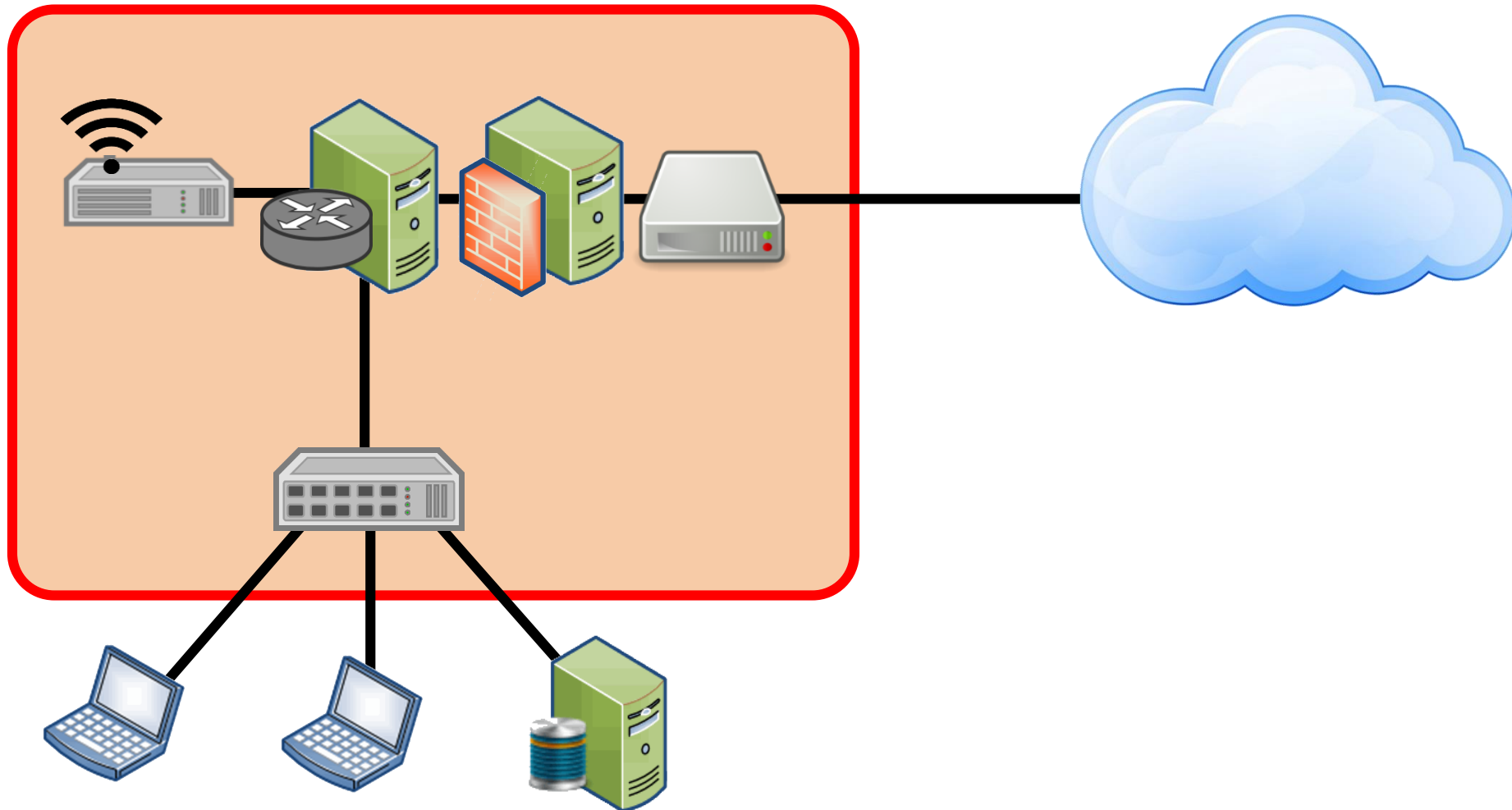
Firewall

- Serves security
- Decides which packets are forwarded based on defined rules



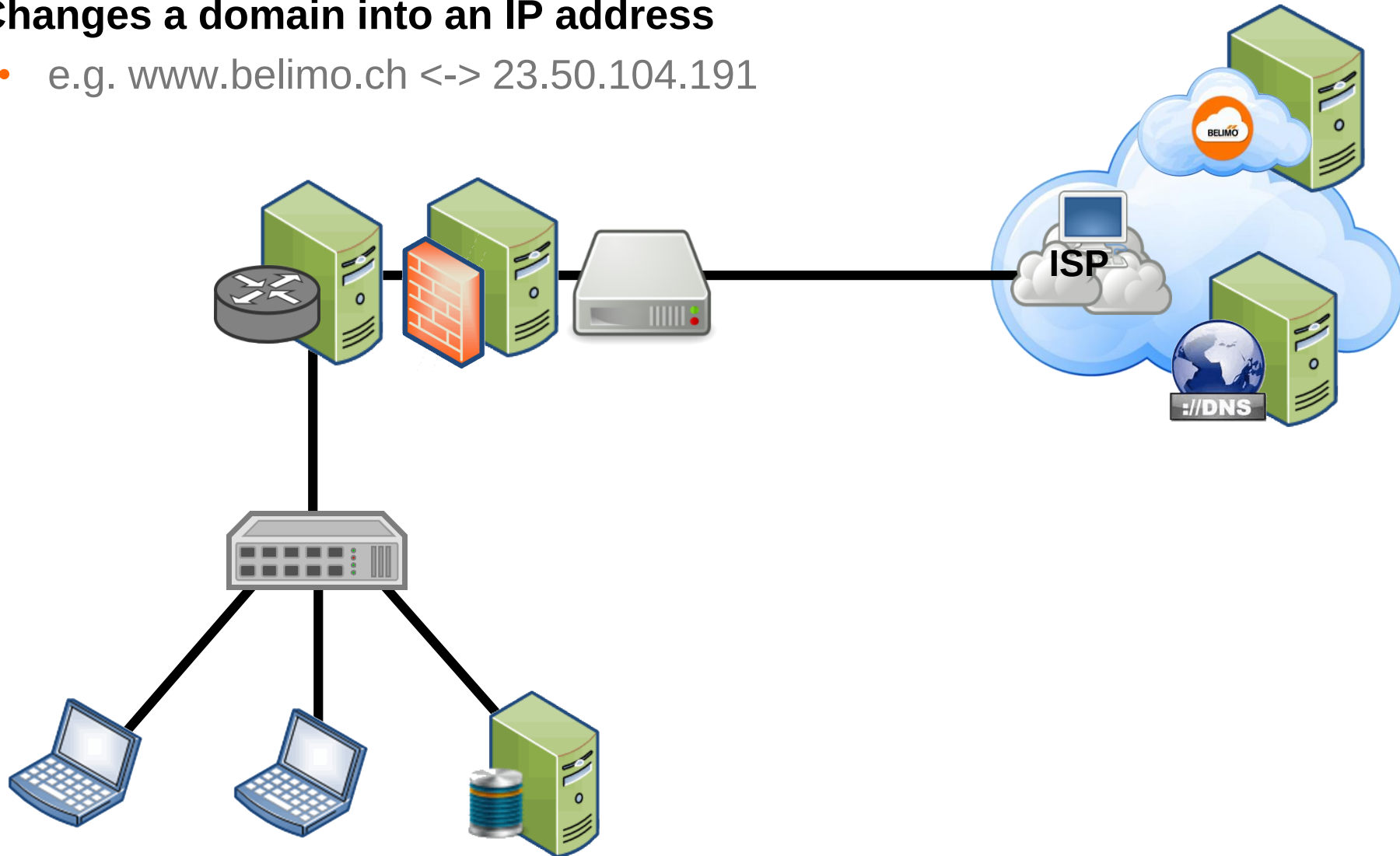
Integrated Access Device (IAD)

- **Combination of various network components**

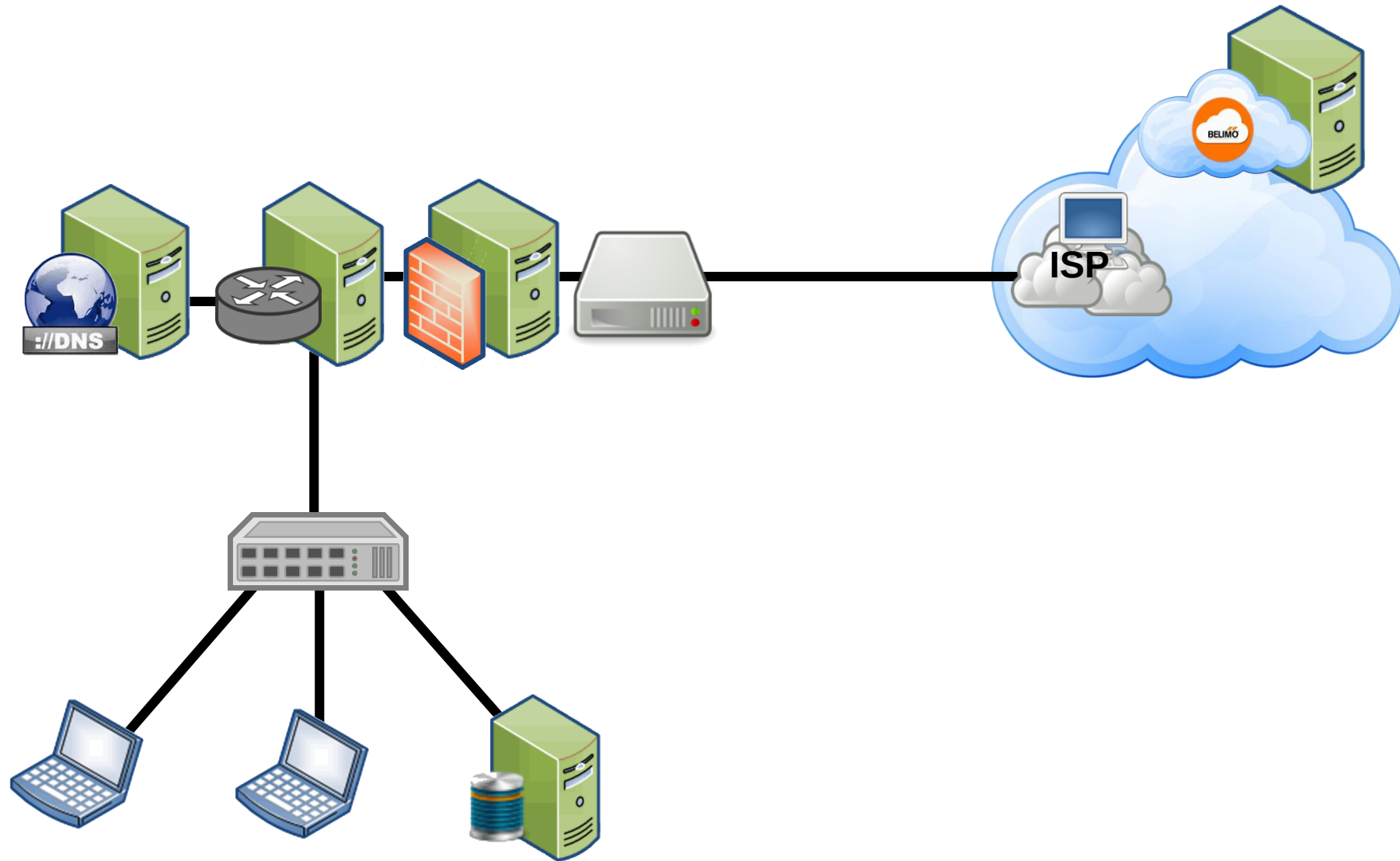


DNS

- **Changes a domain into an IP address**
 - e.g. `www.belimo.ch` <-> `23.50.104.191`



DNS - Local



Recap

- Peer-Peer
- Hub
- Switch
- Router
- ISP
- IAD

Useful links to recap at home:

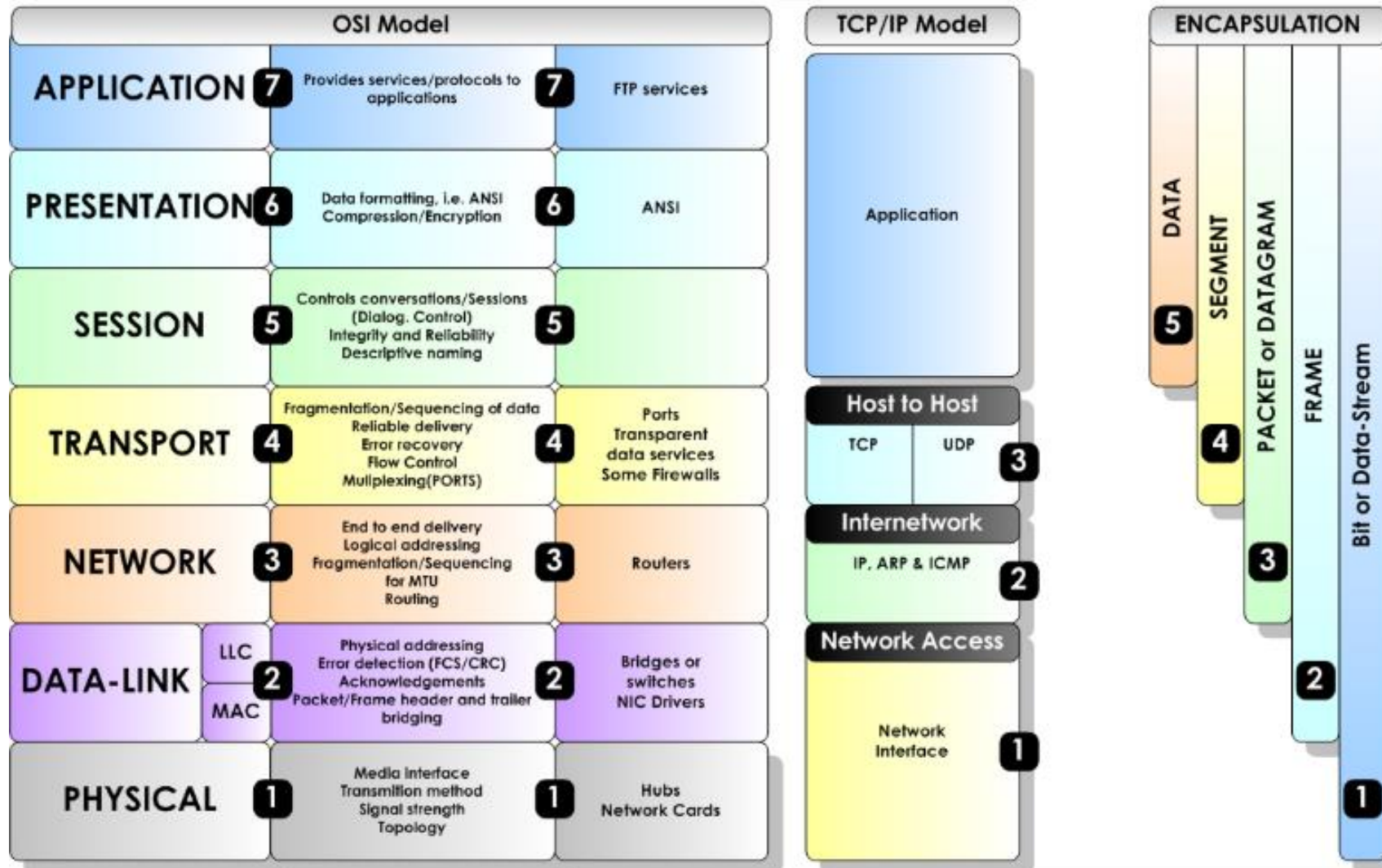
https://www.youtube.com/watch?v=1z0ULvg_pW8&t=23s (7'18'')

Network Models

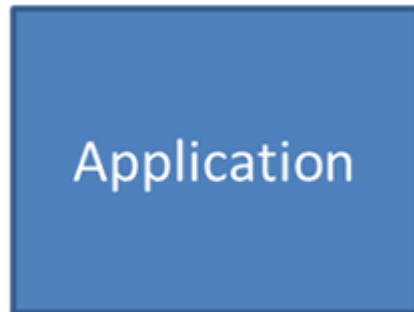
Network Models

The OSI Model (Open Systems Interconnection)

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www.networkstuff.eu



TCP/IP Model



Application layer – Application-specific

e.g. HTTP, FTP, SMTP, POP3, DNS, SSH, Telnet



Transport layer – Reliable transmission

e.g. TCP, UDP



Internet layer – Routing

e.g. IP (IPv4, IPv6), ICMP, IPX



Link layer – Data transmission technologies

e.g. Ethernet, WLAN, Coax, Fibre, ATM, FDDI, PPP, Token Ring



**Application / Presentation / Session layer –
Application-specific (Data)**

z.B. HTTP, FTP, SMTP, POP3, DNS, SSH, Telnet



Transport layer –

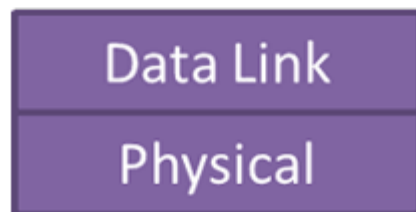
Reliable transmission (Segment)

z.B. TCP, UDP



Network layer – Routing (Packet)

z.B. IP (IPv4, IPv6), ICMP, IPX



Data Link layer –

Techniken der Datenübertragung (Frame)

z.B. Ethernet, WLAN, ATM, FDDI, PPP, Token Ring

Physical layer –

Techniken der Datenübertragung (Bit)

z.B. 1000BASE-T, Token Ring, ARCNET

Components in ISO OSI Modell



Application / Presentation / Session / Transport layer –
Gateway, Content-Switch, Proxy, Layer 4-7-Switch



Network layer –
Router, Layer-3-Switch

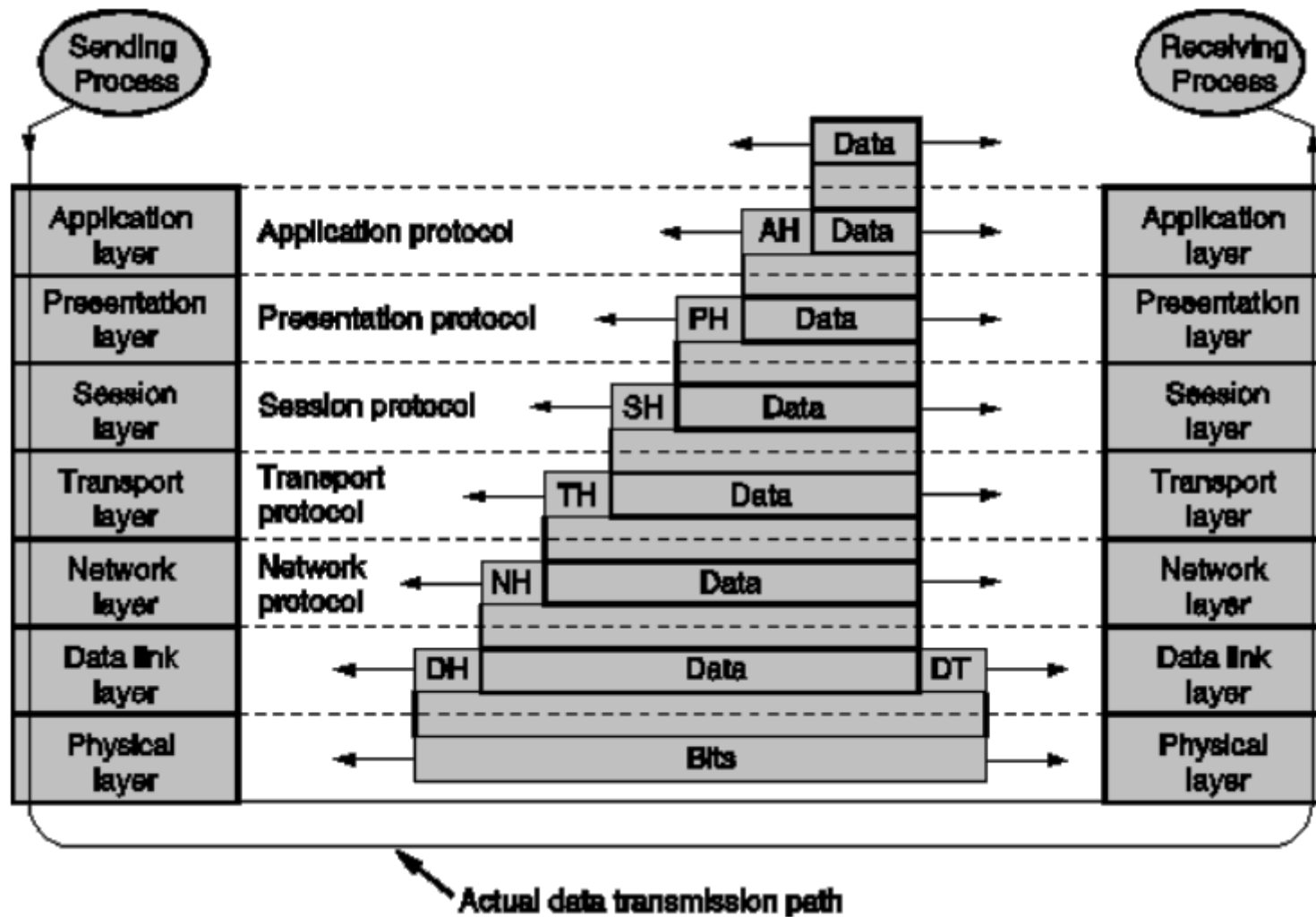


Data Link layer –
Bridge, Layer-2-Switch, Access Point



Physical layer –
network cable, Repeater, Hub

Data Transmission



Recap

Useful links to recap at home:

<https://www.youtube.com/watch?v=HAVcXPI7oUY> (47'02'')



Thank you for your attention